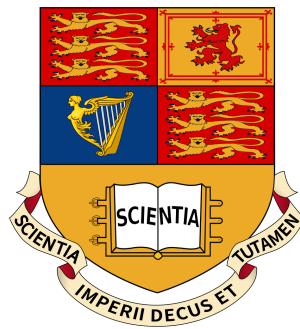


IMPERIAL

MSC SELF-STUDY PROJECT

Surface-GKP Quantum Error Correction

Jude Russell



Imperial College London
Department of Physics
1st March 2026

Abstract

Achieving scalable fault-tolerant quantum computation is currently held back by the large hardware overhead of qubit-based error-correcting codes. This review examines the surface–GKP architecture, which addresses this challenge by concatenating a bosonic Gottesman–Kitaev–Preskill (GKP) inner code with an outer surface code. The GKP code provides hardware efficiency by encoding logical information within a single harmonic oscillator, using its phase space lattice structure to digitise continuous displacement noise into discrete Pauli errors. Applying an outer surface code to correct these residual faults has been shown to reduce the number of physical modes per logical qubit by an order of magnitude compared to standard physical-qubit schemes, provided the effective noise model remains below a finite squeezing threshold. We survey recent experimental progress across trapped-ion, superconducting, and photonic platforms, showing that GKP states have transitioned from theoretical ideas to viable physical qubits amenable to repeated stabiliser-based error correction. Although achieving such high squeezing levels and low-loss entangling gates remains a significant engineering challenge, the surface–GKP architecture represents a promising and increasingly realistic route toward large-scale fault-tolerant quantum computing.

Contents

1	Introduction	1
2	The Gottesman–Kitaev–Preskill Code	1
2.1	Bosonic modes and phase space	1
2.2	Stabiliser definition of the GKP code	2
2.3	Codewords	3
2.4	Finite-energy GKP states	3
3	GKP error correction	5
3.1	Correction protocols	5
3.2	Logical operations and universality	6
4	Experimental realisations of GKP states	7
4.1	Trapped-ion systems	7
4.2	Superconducting systems	8
4.3	Photonic systems	8
5	The surface–GKP code	9
5.1	Concatenation motivation	9
5.2	Structure of the surface–GKP code	10
5.3	Syndrome extraction	11
5.4	Decoding	11
5.5	Fault-tolerance thresholds	12
5.6	Towards an optical surface–GKP code	12
6	Conclusion	13
	References	13

1 Introduction

Quantum computation exploits uniquely quantum-mechanical phenomena to process information in ways that are provably intractable for classical machines. In the circuit model, information is encoded in qubits: two-level quantum systems whose state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (1)$$

evolves under unitary gates and is ultimately read out by measurement. Entanglement between multiple qubits enlarges the accessible Hilbert space exponentially with system size, enabling quantum algorithms for factoring [1], unstructured database search [2], and quantum simulation [3].

Despite this potential, physical platforms on which qubits are currently implemented all share a high susceptibility to errors. Classical computers deal with errors by continuously stabilising bits to discrete 0 or 1 values. This approach is fundamentally incompatible with quantum mechanics, as directly measuring a qubit collapses its superposition, destroying the encoded information. Entangled states are remarkably delicate, and as quantum processors scale in size, decoherence from interaction with the environment becomes increasingly difficult to suppress.

Quantum error correction provides the essential solution to this fragility. Rather than attempting to guard against all environmental interactions, the approach is to encode a single logical qubit into a larger Hilbert space such that errors can be detected and reversed without measuring the encoded information directly. In qubit-based computation, this is achieved using stabiliser codes, where logical states are defined as joint eigenspaces of commuting operators and errors are identified by the syndromes they produce. The threshold theorem guarantees that arbitrarily long quantum computation is possible, provided the error rate remains below a finite value [4].

The surface code has emerged as a leading candidate for large-scale fault tolerance [5]. It arranges qubits on a two-dimensional lattice with local stabiliser measurements, achieving a high error threshold. Nevertheless, it comes with the substantial cost that many physical two-level systems are required to encode and protect a single logical qubit. This motivates the search for more hardware-efficient encodings.

Many experimental platforms naturally provide bosonic modes — harmonic oscillators with infinite-dimensional Hilbert spaces — rather than two-level systems. The Gottesman–Kitaev–Preskill (GKP) code exploits this by encoding a logical qubit into a phase-space lattice of oscillator states [6], converting continuous displacement noise into discrete Pauli errors that higher-level codes can correct. Since finite-energy GKP states cannot provide complete fault tolerance alone, concatenating them with a surface code yields the surface–GKP architecture: the GKP layer digitises continuous errors, and the surface code suppresses the residual discrete faults.

This review develops the theoretical foundations of the GKP code, surveys its experimental realisations, and analyses its integration with surface codes as a practical route towards scalable, fault-tolerant quantum computation.

2 The Gottesman–Kitaev–Preskill Code

2.1 Bosonic modes and phase space

Standard qubits are thought of as two level systems, provided discretely by nature, perhaps as the spin of an electron, or the polarisation of a photon. But there is another way of making a qubit, and that is in the form of a bosonic system. A bosonic system is one with continuous variables (CVs). Examples of bosonic systems are harmonic oscillators, which are ubiquitous in familiar systems, such as an optical field, the motion of a trapped ion, or a microwave cavity. These continuous quantum variable systems come with the advantage of an infinite-dimensional Hilbert space, as opposed to the finite spaces that come with two-level qubits. As will be elaborated on later, this allows for the crucial property of hardware efficiency when making a quantum processor [6, 7].

Bosonic modes are governed by position and momentum operators, defined with the usual creation and annihilation operators as

$$\hat{q} = \frac{(\hat{a}^\dagger + \hat{a})}{\sqrt{2}} \quad \text{and} \quad \hat{p} = \frac{i(\hat{a}^\dagger - \hat{a})}{\sqrt{2}} \quad (2)$$

respectively, where $[\hat{a}, \hat{a}^\dagger] = 1$ resulting in $[\hat{q}, \hat{p}] = i$, where we have set $\hbar = 1$. It is useful to conceptualise bosonic modes as existing in phase space with position and momentum axes, and a complete description of the state can be given by its Wigner function, which is a quasi-probability distribution. The Wigner function encodes all information about the quantum state and can take on negative values. Noise in CV systems appears as small random displacements in phase space, given mathematically by the displacement operator

$$\hat{D}(\alpha) = e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}} = e^{i(\nu \hat{q} - u \hat{p})} \equiv \hat{D}(u, \nu), \quad (3)$$

where the complex amplitude $\alpha = (u + i\nu)/\sqrt{2}$ corresponds to real-valued displacements u and ν in the q and p quadratures, respectively. The operators are generally non-commutative, obeying

$$\hat{D}(\beta)\hat{D}(\alpha) = e^{(\beta\alpha^* - \beta^*\alpha)/2} \hat{D}(\alpha + \beta) = e^{\beta\alpha^* - \beta^*\alpha} \hat{D}(\alpha)\hat{D}(\beta). \quad (4)$$

In terms of the real quadrature displacements $\mathbf{u}_1 = (u_1, \nu_1)$ and $\mathbf{u}_2 = (u_2, \nu_2)$, the exponential factor can be written as

$$\beta\alpha^* - \beta^*\alpha = i(u_2\nu_1 - u_1\nu_2) \equiv i\omega(\mathbf{u}_1, \mathbf{u}_2), \quad (5)$$

so the operator condition becomes

$$\hat{D}(u_2, \nu_2)\hat{D}(u_1, \nu_1) = e^{-i\omega(\mathbf{u}_1, \mathbf{u}_2)} \hat{D}(u_1, \nu_1)\hat{D}(u_2, \nu_2). \quad (6)$$

The symplectic¹ inner product $\omega(\mathbf{u}_1, \mathbf{u}_2)$ is proportional to the area of the shape formed by the displacements in phase space. Operators commute if this area is an integer multiple of 2π in the exponent such that the phase factor is unity, i.e.,

$$\omega(\mathbf{u}_1, \mathbf{u}_2) = 2\pi n, \quad n \in \mathbb{Z}, \quad (7)$$

and anticommute if it is an odd multiple of π ,

$$\omega(\mathbf{u}_1, \mathbf{u}_2) = \pi(2n + 1), \quad n \in \mathbb{Z}. \quad (8)$$

The main aim of the GKP code is to *digitise* these continuous errors [6], converting small phase space displacements into discrete logical errors that can be handled by higher-level error-correction protocols.

2.2 Stabiliser definition of the GKP code

The GKP code is defined within the stabiliser framework by two commuting displacement operators [6, 8]:

$$\hat{S}_q = e^{i2\sqrt{\pi}\hat{q}}, \quad \hat{S}_p = e^{-i2\sqrt{\pi}\hat{p}}. \quad (9)$$

These stabilisers correspond to translations by $2\sqrt{\pi}$ in phase space, and effectively define the periodic boundaries of a square lattice.² The logical code space is then defined as the simultaneous +1 eigenspace of the two stabilisers, resulting in a grid-like structure in phase space periodic in q and p .

¹Coined by Hermann Weyl in 1939 as a Greek calque for “complex” to avoid ambiguity, symplectic refers to the area-preserving geometry of phase space. Mathematically, it defines the symplectic bilinear form $\omega(\mathbf{u}_1, \mathbf{u}_2) = u_2\nu_1 - u_1\nu_2$.

²The choice of $2\sqrt{\pi}$ is not arbitrary. It ensures that logical Pauli operators, defined as half-lattice shifts, satisfy the required anticommutation relations.

This geometric structure allows a finite-dimensional logical subsystem to be embedded within the infinite-dimensional Hilbert space [9].

Logical Pauli operators for the GKP code are defined as half-lattice translations in phase space:

$$\bar{X} = e^{-i\sqrt{\pi}\hat{p}}, \quad \bar{Z} = e^{i\sqrt{\pi}\hat{q}}. \quad (10)$$

These operators satisfy $\bar{X}^2 = \hat{S}_p$ and $\bar{Z}^2 = \hat{S}_q$. They commute with the stabilisers, but anticommute with each other ($\bar{X}\bar{Z} = -\bar{Z}\bar{X}$) as required for a logical qubit. A logical Pauli error therefore corresponds to a shift of $\sqrt{\pi}$ in phase space.

2.3 Codewords

The ideal computational basis states are defined as infinite superpositions of Dirac delta peaks [6]. They are given in the position quadrature by

$$|0_L\rangle = \sum_{n \in \mathbb{Z}} |q = 2n\sqrt{\pi}\rangle, \quad (11)$$

$$|1_L\rangle = \sum_{n \in \mathbb{Z}} |q = (2n+1)\sqrt{\pi}\rangle. \quad (12)$$

In this basis, $|0_L\rangle$ and $|1_L\rangle$ consist of peaks at even and odd multiples of $\sqrt{\pi}$, respectively. The complementary basis states $|\pm_L\rangle \equiv (|0_L\rangle \pm |1_L\rangle)/\sqrt{2}$ also have this ‘comb’ structure in the momentum quadrature and are given by

$$|+_L\rangle = \sum_{n \in \mathbb{Z}} |p = 2n\sqrt{\pi}\rangle, \quad (13)$$

$$|-_L\rangle = \sum_{n \in \mathbb{Z}} |p = (2n+1)\sqrt{\pi}\rangle. \quad (14)$$

Thus GKP states are lattices in phase space given by simultaneous eigenstates of two commuting displacement operators. The standard GKP lattice is square, which will be assumed throughout this review, but alternative geometries such as rectangular or hexagonal³ are also common [10], and example structures are shown in Fig. 1.

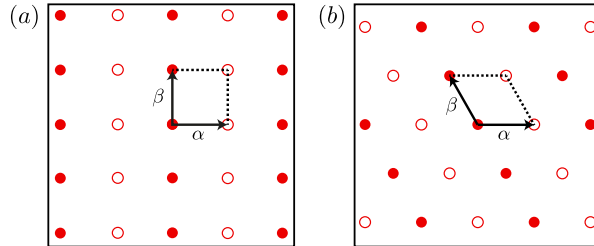


Figure 1. Phase space representation of (a) square and (b) hexagonal GKP lattices. The logical codewords $|0_L\rangle$ (filled circles) and $|1_L\rangle$ (open circles) form interleaved grids, where the separation between identical logical states corresponds to a full lattice translation (stabiliser) and the separation between different logical states corresponds to a half-lattice translation (logical Pauli operator). Reproduced from [10].

2.4 Finite-energy GKP states

The ideal GKP codewords are Dirac comb states consisting of infinitely sharp peaks in phase space. Such states are unphysical because they would require infinite squeezing. Squeezing refers to the process of reducing the uncertainty in one quadrature of a harmonic oscillator below its vacuum

³The smallest uncorrectable displacement in a hexagonal lattice is slightly larger than in a square lattice, by a factor of $(2/\sqrt{3})^{1/2} \approx 1.0746$, because the hexagonal lattice packs lattice points more efficiently in phase space [6].

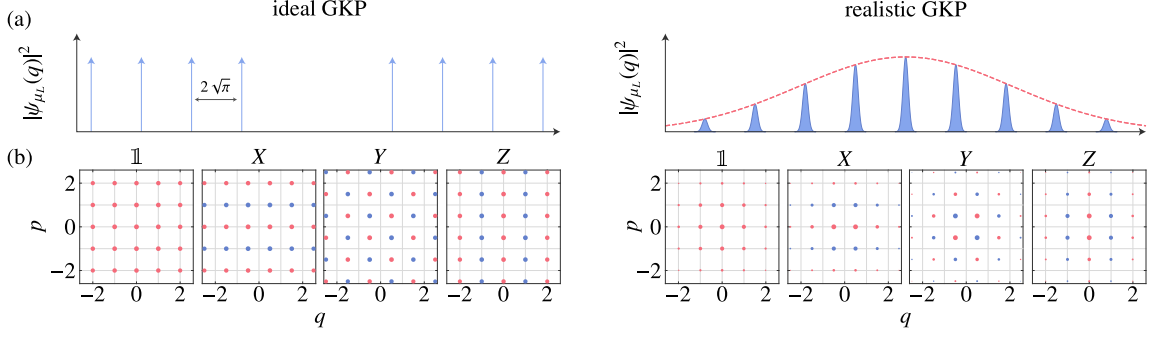


Figure 2. Comparison between ideal and finite-energy GKP states. (a) Position-space probability density $|\psi_{\mu_L}(q)|^2$ for a logical GKP basis state. In the ideal case, the codewords form an infinitely sharp Dirac-comb lattice. For finite energy, each delta peak is broadened into a narrow Gaussian and distant peaks are suppressed by a broad Gaussian envelope (dashed), resulting in a normalisable state with finite mean energy. (b) Phase space representations of the Wigner functions for the logical Pauli operators I , X , Y , and Z . Red (blue) indicates positive (negative) regions, with marker size proportional to the magnitude of each lattice peak. Adapted from [11].

level, at the expense of increasing the uncertainty in the other quadrature in accordance with the uncertainty principle. In the limit of infinite squeezing, the mean photon number, and therefore the energy, diverges, rendering the ideal codewords physically unattainable. Realistic implementations must instead use finitely squeezed approximations in which each Dirac delta peak is replaced by a narrow Gaussian of finite width [6].

Following the formulation in [6, 9], an approximate GKP state can be obtained by applying a Gaussian superposition of displacements to an ideal codeword:

$$|\psi_\Delta\rangle = E_\Delta |\psi_{\text{ideal}}\rangle, \quad (15)$$

where

$$E_\Delta = \frac{1}{\sqrt{\pi\Delta^2}} \int_{\mathbb{R}^2} du dv \exp\left(-\frac{u^2 + v^2}{2\Delta^2}\right) \exp(-iu\hat{p} + iv\hat{q}). \quad (16)$$

Here $\exp(-iu\hat{p} + iv\hat{q})$ is the displacement operator that shifts $\hat{q} \rightarrow \hat{q} + v$ and $\hat{p} \rightarrow \hat{p} + u$. The parameter $\Delta > 0$ characterises the amount of squeezing. Physically, this means that instead of an ideal lattice state with perfectly sharp peaks, we consider a superposition of lattice states that are slightly displaced in phase space. The displacement amplitudes (u, v) are drawn from an isotropic Gaussian distribution with variance Δ^2 . In this way, Δ quantifies the intrinsic uncertainty in both quadratures around each lattice point. The qualitative difference between ideal and finitely squeezed codewords is illustrated in Fig. 2. Panel (a) shows the position-space probability density $|\psi_{\mu_L}(q)|^2 = |\langle q|\mu_L\rangle|^2$ for a logical GKP basis state $|\mu_L\rangle$ with $\mu \in \{0, 1\}$. In the ideal case this distribution forms a Dirac comb with lattice spacing $2\sqrt{\pi}$, whereas finite squeezing replaces each delta peak by a Gaussian of width $\sim \Delta$ and introduces a global Gaussian envelope.

For small Δ , one may approximate

$$E_\Delta \approx 2\sqrt{\pi\Delta^2} D_\Delta, \quad D_\Delta \equiv \exp(-\Delta^2 \hat{n}), \quad (17)$$

with the photon number operator $\hat{n} = \hat{a}^\dagger \hat{a}$. This D -approximation makes explicit the origin of the broad Gaussian envelope. Another small Δ approximation makes clear the characteristic comb-with-envelope form (the F -approximation) as shown in Fig. 2:

$$E_\Delta |0_\Delta\rangle \approx F_\Delta |0_\Delta\rangle \propto \int_{\mathbb{R}} dq \sum_{k \in \mathbb{Z}} \underbrace{e^{-2\pi\Delta^2 k^2}}_{\text{envelope}} \underbrace{e^{-\frac{1}{2\Delta^2}(q-2k\sqrt{\pi})^2}}_{\text{comb}} |q\rangle. \quad (18)$$

In phase space, the Wigner function retains the periodic grid structure of the ideal code but with peaks broadened to width $\sim \Delta$, while the overall lattice is weighted by a Gaussian envelope $e^{-2\pi\Delta^2 k^2}$ ensuring normalisability and finite energy.

For the E_Δ approximation, the mean photon number satisfies $\bar{n} \approx \frac{1}{2\Delta^2} - \frac{1}{2}$ [9], so that $\bar{n} \sim 1/(2\Delta^2)$ for small Δ . Hence, as $\Delta \rightarrow 0$, the mean energy diverges and the ideal codewords are recovered. It is customary to express the quality of an approximate GKP state in terms of squeezing S in decibels (see §5.5),

$$S = -10 \log_{10}(\Delta^2). \quad (19)$$

A natural noise model for bosonic encodings is the Gaussian displacement channel $\mathcal{N}(\rho)$ with standard deviation σ :

$$\mathcal{N}(\rho) = \frac{1}{2\pi\sigma^2} \int_{-\infty}^{\infty} du \int_{-\infty}^{\infty} dv e^{-(u^2+v^2)/2\sigma^2} e^{-iu\hat{p}+iv\hat{q}} \rho e^{iu\hat{p}-iv\hat{q}}, \quad (20)$$

where ρ is the density matrix for a single-mode [9].

In these terms, the preparation of an approximate GKP state via E_Δ can itself be interpreted as the addition of an intrinsic noise source of variance Δ^2 to an ideal lattice state. When external Gaussian displacement noise of variance σ^2 is introduced as well, the total variance adds linearly:

$$\Delta_{\text{eff}}^2 = \Delta^2 + \sigma^2. \quad (21)$$

Logical errors occur when shifts move the state outside the correctable region of the square lattice, i.e., beyond half the logical operator separation ($\sqrt{\pi}/2$) in either quadrature [6]. Thus, the performance of the code is governed by the relation between the effective variance Δ_{eff}^2 and the lattice spacing set by the stabilisers $\hat{S}_q$ and $\hat{S}_p$. In this way, finite-energy GKP states can be consistently viewed as ideal grid states subjected to Gaussian displacement noise characterised by the squeezing parameter Δ .

3 GKP error correction

3.1 Correction protocols

We have established that noise in CV systems consists of small displacements in position or momentum that move the state away from the lattice points. In order to correct these errors, the stabilisers of the GKP code (Eq. (9)) are measured non-destructively, which is equivalent to measuring the modular quadratures $\hat{q} \pmod{\sqrt{\pi}}$ and $\hat{p} \pmod{\sqrt{\pi}}$ [6]. These measurements, or syndromes, project the state into a specific error subspace without collapsing the encoded logical information. Once the small displacement from the nearest lattice point is identified, the state is restored to the code lattice via corrective phase space shifts, effectively discretising continuous noise into residual logical errors [12].

Two primary protocols are commonly employed for GKP error correction, as illustrated in Fig. 3 and Fig. 4. The first, *Steane-type correction* [9, 13], involves the direct coupling of a noisy data state $|\psi\rangle$ to two GKP ancilla states — typically $|0_L\rangle$ (to extract q -shifts) and $|+L\rangle$ (to extract p -shifts). The interaction is mediated by the SUM gate, $\text{SUM}_{D \rightarrow A} = e^{-i\hat{q}_D \hat{p}_A}$, the CV analog of the CNOT gate [6, 13]. In the Heisenberg picture, the quadratures transform as

$$\hat{q}_A \rightarrow \hat{q}_A + \hat{q}_D, \quad \hat{p}_D \rightarrow \hat{p}_D - \hat{p}_A, \quad (22)$$

where the subscripts D and A denote the data and ancilla qubits, respectively.

Alternatively, *Knill-type correction* [14] implements correction by teleporting the logical state through a GKP Bell pair. This is highly advantageous for optical platforms, as it delegates complexity to the offline preparation of resource states, allowing online correction to be performed with passive linear optics [12, 14]. The required resource is a Bell state projected into the GKP subspace,

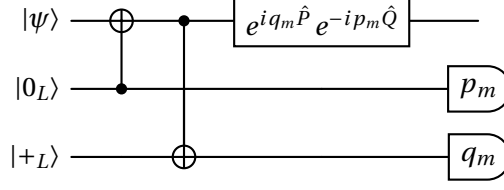


Figure 3. Steane-type GKP error correction circuit. The noisy data state $|\psi\rangle$ is coupled to two GKP ancilla states ($|0_L\rangle$ and $|+_L\rangle$) via SUM gates (represented as controlled-NOT interactions). Homodyne measurements of the ancillas give the displacement syndromes p_m and q_m . These outcomes determine the adaptive feedback displacements $e^{iq_m \hat{P}}$ and $e^{-ip_m \hat{Q}}$ required to shift the data state back to the nearest code lattice point.

often generated by interfering two “qunaught” states (GKP states with no logical information) on a 50:50 beam splitter. This results in a state where the modes are correlated according to the GKP stabiliser conditions:

$$\hat{q}_1 - \hat{q}_2 \equiv 0 \pmod{\sqrt{\pi}}, \quad \hat{p}_1 + \hat{p}_2 \equiv 0 \pmod{\sqrt{\pi}}. \quad (23)$$

As illustrated in Fig. 4, error correction proceeds by interfering the noisy data mode with the first half of the Bell pair. Subsequent homodyne measurements of the $\hat{q}$ and $\hat{p}$ quadratures constitute a Bell-basis measurement, giving the syndrome outcomes q_m and p_m . The logical information is thereby teleported to the second member of the Bell pair, which acts as a fresh physical carrier. This ‘mode-reset’ effectively discards the original noisy mode in favour of a fresh, GKP-protected ancilla [12, 14].

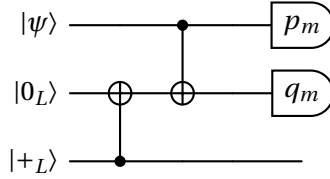


Figure 4. Knill-type GKP error correction circuit. The circuit first prepares a GKP Bell pair by entangling two ancilla modes ($|0_L\rangle$ and $|+_L\rangle$) using a SUM gate. The noisy data state $|\psi\rangle$ then interacts with the upper half of this Bell pair. Subsequent homodyne measurements extract the momentum and position syndromes (p_m, q_m), which determine the shift required to return the state to the lattice center. The logical information is teleported to the lower mode of the Bell pair, effectively transferring it to a fresh oscillator and preventing error accumulation on the original physical mode.

In either method, error correction succeeds provided the displacement is less than $\sqrt{\pi}/2$, or half a logical operator separation [6]. If the shift exceeds this threshold, the state is mapped to an incorrect neighbouring lattice point, resulting in a logical Pauli error. Such errors will be dealt with in §5.

3.2 Logical operations and universality

A primary advantage of the GKP encoding is that the entire Clifford group⁴ of quantum operations can be implemented using only Gaussian operations [15]. In optical and microwave platforms, this enables logical processing using linear optics, squeezing, and homodyne detection [6].

Since the GKP code space is defined by discrete translations in phase space, the logical Pauli operators $\tilde{X}$ and $\tilde{Z}$ are naturally realised as displacement operators in momentum and position, respectively. More generally, the entire Clifford group corresponds to symplectic transformations that

⁴The Clifford group consists of unitaries that map Pauli operators to Pauli operators, generated by the Hadamard, phase, and CNOT gates.

preserve both the GKP lattice structure and the canonical commutation relations. Familiar gates have simple geometric interpretations: the Hadamard is a $\pi/2$ phase-space rotation exchanging $\hat{q}$ and $\hat{p}$, the phase gate is a shear, and the CNOT is realised through a SUM interaction between modes. A key feature of these Gaussian operations is that they map small displacements to small displacements, so errors remain correctable after a gate is applied [6, 8, 9].

However, Gaussian operations alone are not sufficient for universal quantum computation. Circuits composed of Gaussian states and Gaussian operations can be efficiently simulated on a classical computer whenever the Wigner function remains positive [16]. Consequently, non-Gaussian resources are required to achieve quantum computational advantage [6, 15].

In continuous-variable architectures, this requirement can be met through non-Gaussian resource states. One candidate is the cubic phase state $|\gamma\rangle = e^{i\gamma\hat{q}^3}|0\rangle$ [6],⁵ though this approach has been shown to perform poorly even at arbitrary squeezing [17]. A more practical route is magic-state injection [18], in which noisy non-Gaussian ancillas are purified using Clifford operations and GKP error correction, enabling implementation of a non-Clifford gate such as the T gate on the data mode. Together with the Clifford operations already available, this completes a universal gate set, confirming that the surface-GKP architecture can in principle support universal fault-tolerant quantum computation. Achieving universality therefore ultimately relies on the ability to generate high-fidelity non-Gaussian states and to realise the nonlinear interactions needed to prepare them [6, 15].

4 Experimental realisations of GKP states

The experimental development of the GKP code has progressed along three primary physical platforms: trapped ions, superconducting circuit quantum electrodynamics (cQED), and photonic systems. Each architecture implements the same stabiliser structure of periodic phase space translations of a harmonic oscillator, but differs fundamentally in how modular quadrature measurements, feed-back, and state preparation are realised. They also differ in their trade-offs between squeezing, coherence time, and scalability [9].

4.1 Trapped-ion systems

The first full experimental realisation of a GKP qubit was achieved by Flühmann *et al.* in 2019 [19]. In this work, the bosonic mode was the motion of a single $^{40}\text{Ca}^+$ ion confined in a linear Paul trap, while two internal electronic states served as an ancillary qubit enabling entangling operations and measurement.

Optical dipole forces entangled the ion's motional state with its internal electronic state. Measuring the latter via fluorescence projected the motion into superpositions of displaced squeezed states. Iterating this process built up the grid structure in phase space, producing GKP states with approximately 10 dB of squeezing [19].

A key limitation of this specific implementation was its reliance on post-selection: runs that produced unwanted measurement outcomes were simply discarded in order to improve the fidelity of the prepared states. Although this approach yielded high-quality logical states, it does not scale well. Fault-tolerant operation requires stabiliser measurements that can be performed deterministically and repeatedly, without rejecting data.

Logical Pauli operations were implemented as carefully calibrated displacements in phase space. Arbitrary single-qubit rotations were achieved through gate teleportation using the internal-state ancilla, thereby realising a universal set of single-logical-qubit gates. The reported process fidelities were about 97 % for Pauli gates and about 87 % for teleported $\pi/2$ and $\pi/4$ rotations. The average logical state preparation and measurement fidelity was 87.3 %, with coherence times of order 3.7 ms, limited mainly by finite squeezing and motional dephasing.

⁵ $e^{i\gamma\hat{q}^3}$ is applied to a vacuum or squeezed state, where the real parameter γ sets the strength of the cubic nonlinearity.

Overall, the experiment demonstrated encoding, control, and characterisation of a bosonic GKP logical qubit. It showed that modular-variable measurements provide a viable method for preparing grid states. At the same time, the reliance on post-selection and the lack of repeated, real-time error correction highlighted the need for architectures that can perform stabiliser measurements and feedback autonomously.

4.2 Superconducting systems

Soon afterwards, Campagne-Ibarcq *et al.* [20] reported the demonstration of repeated, measurement-based quantum error correction of a GKP qubit without relying on post-selection. In their architecture, the bosonic mode is a three-dimensional microwave cavity that serves as the logical qubit, coupled to the Pauli operator σ_z of a superconducting two-level ancilla called a transmon qubit. The GKP stabilisers were measured via a conditional displacement interaction

$$U_{CD}(\beta) = e^{i(-\text{Re}(\beta) p + \text{Im}(\beta) q) \frac{\sigma_z}{2}}. \quad (24)$$

This interaction was implemented via the dispersive coupling between the superconducting cavity and the transmon, combined with microwave drives that engineer a conditional displacement of the cavity state dependent on the transmon's internal state. When embedded in a Ramsey sequence on the transmon, this maps the expectation value of a cavity displacement operator onto a Pauli observable of the transmon, enabling measurement of the modular quadratures in phase space. The resulting binary outcomes are processed in real time to apply corrective displacements, steering the cavity state toward the GKP code space.

Unlike previous trapped-ion experiments that relied on post-selection, stabiliser measurements here were repeated deterministically, without discarding experimental runs. Starting from the vacuum state, iterative rounds of stabiliser measurement and feedback prepared approximate square-lattice GKP states. The measured peak squeezing oscillated between 7.4 dB and 9.5 dB across stabiliser measurement rounds, reflecting the fact that each round preferentially squeezes one quadrature at the expense of the other.

Continuous stabiliser monitoring extended the logical coherence time. For the square-lattice GKP code, the reported logical lifetimes were $T_X = T_Z \approx 275 \mu\text{s}$ and $T_Y \approx 160 \mu\text{s}$, in agreement with numerical simulations [20]. A hexagonal-lattice encoding was also realised, yielding nearly isotropic logical decay with a lifetime of approximately $205 \mu\text{s}$.

This work represents the first deterministic, repeated implementation of GKP quantum error correction, stabilising a qubit without post-selection. The long intrinsic cavity lifetime ($\sim 245 \mu\text{s}$) and high-fidelity Gaussian control position superconducting cavities as a leading platform for scalable bosonic fault-tolerant quantum computing.

4.3 Photonic systems

Photonic systems provide a natural CV platform, with the quadratures of propagating light modes forming bosonic degrees of freedom [21]. Deterministic Gaussian operations, such as squeezing and linear-optical transformations, can be implemented with high fidelity using standard optical components, including near-ideal beam splitters and high-efficiency homodyne detectors [21, 22]. However, GKP states are intrinsically non-Gaussian, requiring non-Gaussian resources for their preparation [15, 23].

Early proposals for optical GKP state generation relied on measurement-induced nonlinearities, where squeezed states were interfered on beam splitters and conditioned on photon-number-resolving detection events to herald grid-like states [16]. These schemes were limited in success rate by optical loss, detector inefficiency, and the probabilistic nature of the measurement.

Recently, Konno *et al.* [24] demonstrated the first generation of approximate square-lattice GKP

states in propagating optical modes. Their method used interference of optical cat states⁶ followed by homodyne measurement to project the output mode into a multi-peaked GKP-like grid. The conditional measurement effectively generates the peak structure, while additional Gaussian operations, such as squeezing, refine the peak widths and spacing. For ideal GKP states, the stabiliser expectation values $\langle \hat{S}_p \rangle$ and $\langle \hat{S}_q \rangle$ are both equal to unity. In practice, finite squeezing broadens the peaks of the phase space lattice, reducing these expectation values below 1. This procedure realises states with average stabiliser values

$$\langle \hat{S}_p \rangle \approx 0.170 \pm 0.003, \quad \langle \hat{S}_q \rangle \approx 0.216 \pm 0.006, \quad (25)$$

where the asymmetry between the two values reflects unequal squeezing in the p and q quadratures of the prepared states. Both values are significantly below the ideal value of 1 due to the limited squeezing achievable in current optical implementations, but nonetheless confirm the presence of the underlying grid structure and demonstrate logical encoding.

The photonic platform offers key advantages for scalability, high-speed operation, and natural integration with fibre-based telecommunication infrastructure, making it well-suited to *distributed quantum computing* architectures [25]. Unlike stationary qubit implementations, propagating optical GKP states can naturally support multiplexing in time or frequency, enabling large-scale multimode operations. Nevertheless, challenges remain. Optical loss, limited squeezing, and the probabilistic nature of cat-state-based preparation constrain the fidelity and determinism of current protocols. Future improvements could leverage high-rate generation of squeezed cat states and feedforward-based displacement⁷ to move toward semi-deterministic preparation [24, 26].

Taken together, experimental progress shows that the GKP code has moved beyond purely theoretical proposals. Trapped ions established full logical control, superconducting cavities enabled repeated stabiliser-based quantum error correction, and photonic platforms have now realised propagating grid states suitable for large-scale multimode architectures. Across all platforms, the central remaining challenge is higher squeezing, lower loss, and deterministic stabiliser measurements.

5 The surface-GKP code

5.1 Concatenation motivation

The GKP code provides a hardware-efficient encoding of a qubit into a single harmonic oscillator mode [6]. Its stabilisers define a square lattice in phase space, and error correction consists of measuring the quadratures modulo $\sqrt{\pi}$ and shifting the state back to the nearest lattice point. Within the fundamental cell of the lattice, small displacement errors are corrected exactly. If the total displacement exceeds half the logical operator separation in either quadrature, the recovery procedure maps the state to a neighbouring lattice point, thereby inducing a logical Pauli error [12].

The key role of the GKP code is to convert continuous Gaussian displacement noise into discrete logical X and Z errors. The oscillator may wander continuously in phase space, but the syndrome measurement extracts a value modulo $\sqrt{\pi}$ and rounds it to the nearest lattice site. Small shifts are corrected; large shifts are interpreted as logical flips.

This discretisation is powerful, but it is not by itself sufficient for scalable quantum computation. As discussed in §2.4, finite-energy GKP states possess intrinsic quadrature uncertainty Δ^2 , and realistic devices introduce additional Gaussian displacement noise σ^2 . The effective variance (see Eq. (21)) therefore remains nonzero. The probability that a shift exceeds $\sqrt{\pi}/2$ is exponentially small in $1/\Delta_{\text{eff}}^2$, but never vanishes. A single GKP qubit therefore behaves as a noisy discrete qubit with a finite Pauli error probability.

⁶A cat state is a superposition of two coherent states with opposite phases, $|\alpha\rangle + |-\alpha\rangle$. When interfered, these states can form the multi-peaked grid structure of a GKP state — a process colloquially known as “cat breeding.”

⁷Feedforward refers to the real-time application of a corrective phase space displacement conditioned on a prior measurement outcome.

As required by the general theory of fault tolerance [27], a viable encoding must ensure that single faults produce at most one correctable error per block, and that the logical error rate decreases with code size. The single-mode GKP code cannot satisfy these requirements on its own. Large displacement events remain possible, and displacement errors can propagate between modes during entangling gates, leading to correlated logical errors [8].

The natural solution is concatenation. The GKP code forms the inner layer, converting physically natural displacement noise into discrete Pauli errors. An outer stabiliser code, most commonly the surface code [5], then corrects these residual Pauli errors. Provided the effective Pauli error rate p after GKP correction lies below the threshold p_{th} of the outer code, the logical failure probability p_L decreases rapidly with the surface-code distance (lattice length) d . Specifically, for $p < p_{\text{th}}$, the logical error rate follows the power law

$$p_L \approx 0.03 \left(\frac{p}{p_{\text{th}}} \right)^{d_e}, \quad (26)$$

where $d_e = \lfloor (d+1)/2 \rfloor$ is the error dimension [5]. Since d_e grows with d , the logical error rate is suppressed super-exponentially as the code distance increases, provided the physical error rate remains below threshold. In this way, the concatenated scheme breaks the problem into two steps: first correcting small displacement errors at the oscillator level, and then correcting the remaining Pauli errors at the qubit level.

5.2 Structure of the surface-GKP code

The surface code is a topological stabiliser quantum error-correcting code [5]. It was originally introduced with periodic boundary conditions as the toric code, and later adapted to planar geometries. A further refinement leads to the *rotated* surface code, in which the lattice is reoriented so that the same code distance d can be achieved using roughly half as many physical qubits as the unrotated version [5]. In its qubit formulation, the code is defined on a two-dimensional square lattice with d^2 data qubits and $d^2 - 1$ syndrome qubits. In the concatenated surface-GKP code, each data qubit is replaced by a GKP-encoded oscillator mode with quadratures $\hat{q}_j$ and $\hat{p}_j$. The layout of the concatenated surface-GKP code for representative distances $d = 3$ and $d = 5$ is shown in Fig. 5.

The logical Pauli operators of the j th GKP qubit are

$$\hat{Z}_j = e^{i\sqrt{\pi}\hat{q}_j}, \quad \hat{X}_j = e^{-i\sqrt{\pi}\hat{p}_j}. \quad (27)$$

The lattice of the surface code is built from plaquettes (faces) and vertices (nodes). Each plaquette hosts a Z-type stabiliser and each vertex hosts an X-type stabiliser, with both stabilisers acting on the four qubits surrounding the plaquette or connected to the vertex. These four-body stabilisers enforce local parity constraints throughout the bulk of the lattice, where every stabiliser involves exactly four qubits. The surface code is defined by four-body stabilisers in the bulk [5],

$$\hat{S}_Z(f) = \prod_{j \in \partial f} \hat{Z}_j, \quad \hat{S}_X(v) = \prod_{j \in \partial v} \hat{X}_j, \quad (28)$$

where f labels plaquettes and v labels vertices. These operators enforce parity constraints among neighbouring encoded qubits.

Because the logical Pauli operators of a GKP qubit are implemented as displacement operations in phase space, each surface-code stabiliser (being a product of Pauli operators) becomes a combined displacement acting across several oscillator modes. In physical terms, the outer surface code therefore imposes joint constraints on multiple oscillators at once. Rather than each mode behaving independently, the logical information is constrained jointly through phase space conditions that couple neighbouring oscillators.

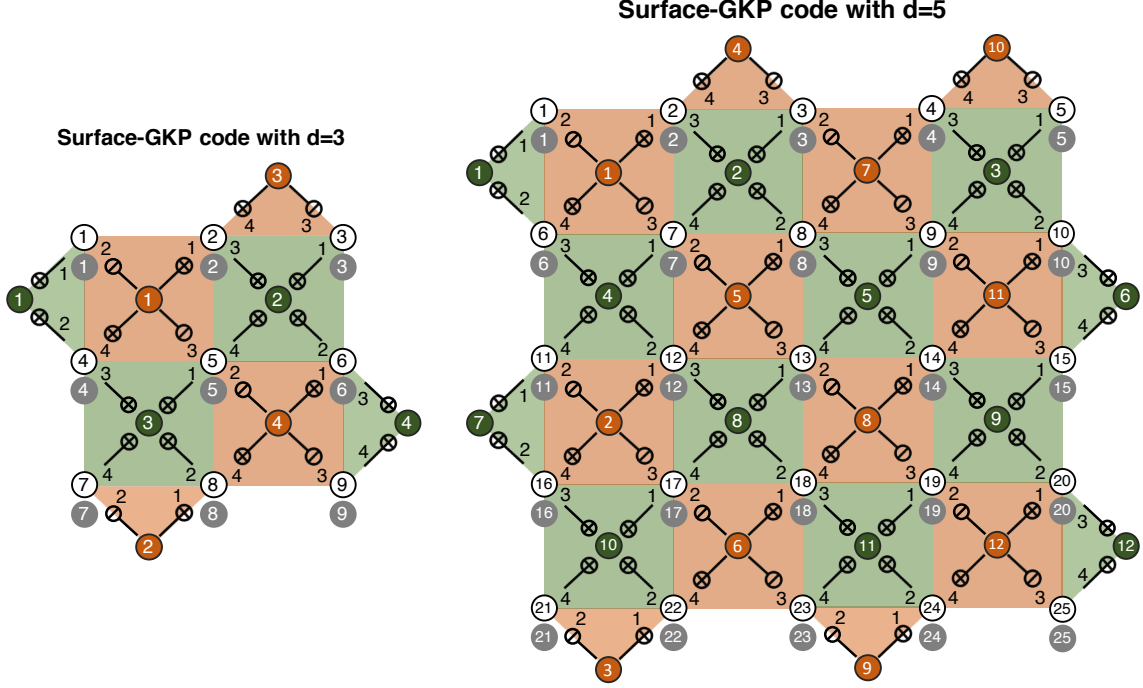


Figure 5. Structure of the concatenated surface-GKP code for code distances $d = 3$ and $d = 5$. White circles denote data GKP modes that encode the logical qubits of the surface code. Grey circles indicate auxiliary GKP modes used to measure the single-mode GKP stabilisers associated with each data oscillator. Green and orange circles represent the syndrome GKP modes used to measure the Z -type (plaquette) and X -type (vertex) surface-code stabilisers, respectively. For a distance- d rotated surface code, there are d^2 data GKP modes and $(d^2 - 1)/2$ syndrome modes of each stabiliser type. The geometry illustrates how local four-body surface-code stabilisers couple neighbouring encoded oscillators through joint displacement constraints in phase space. Reproduced from [8].

5.3 Syndrome extraction

At the concatenated level, we measure the surface-code stabilisers by mapping the parity of multiple GKP data qubits onto an ancilla (syndrome) oscillator. This is implemented using sequences of SUM gates (defined in §3.1), the CV analog of the CNOT gate. The transformation in Eq. (22) shows how noise moves through the circuit. A position displacement on the data mode (the control) propagates forward to the ancilla, while a momentum displacement on the ancilla (the target) propagates backward to the data. This mirrors how X and Z errors behave in a standard qubit surface code: q -shifts act as bit-flips and p -shifts act as phase-flips [6].

In a full stabiliser measurement cycle, each data oscillator is coupled to a syndrome oscillator. After the SUM gates, we measure the syndrome mode using homodyne detection. This measurement result is processed modulo $\sqrt{\pi}$ to give an ‘analog’ remainder $z \in [-\sqrt{\pi}/2, \sqrt{\pi}/2)$. This remainder tells us how far the state has drifted from the ideal lattice point.

In real experiments, we also have to deal with photon loss and gate imperfections. These are usually modelled as Gaussian displacement noise, which adds up as the state moves through the circuit. For the surface code to suppress these errors, the total noise must be kept below a specific threshold, typically cited as 18 dB to 19 dB of effective squeezing in circuit-level studies [8].

5.4 Decoding

A central advantage of the surface-GKP architecture is that GKP measurements produce continuous-valued outcomes, and that analog information z provides a measure of error reliability. If the underlying noise is Gaussian, the likelihood of a logical error increases as $|z|$ approaches the boundary $\sqrt{\pi}/2$.

Instead of converting this information into a simple binary decision, one can retain the full analog value and assign a reliability weight based on the conditional probability of error [28].

Surface-code decoding is then performed using minimum-weight perfect matching (MWPM) on a three-dimensional spacetime lattice [5]. Incorporating the analog reliability information, a form of ‘soft-decoding’, significantly improves performance compared to ‘hard-decision decoding’ [8]. Physically, the GKP layer provides graded information about the quality of each physical qubit, allowing the outer code to focus correction resources on the least reliable qubits.

5.5 Fault-tolerance thresholds

The performance of the surface–GKP code is usually measured by the squeezing threshold — the minimum squeezing required for error correction to reduce the logical error rate. If the squeezing is below this threshold, adding more physical modes (increasing the code distance d) actually makes the logical error rate worse rather than better.

Under a simplified “phenomenological” noise model, where we only consider the finite squeezing of the GKP states themselves, this threshold has a value of approximately 11.2 dB⁸ ($\sigma \approx 0.194$) [8]. However, we can consider a more realistic “circuit-level” noise model where we account for the fact that every SUM gate, every idle period, and every measurement adds noise through photon loss and the finite squeezing of the ancilla states used for syndrome extraction. Because noise accumulates at every step of the circuit, the threshold increases significantly to approximately 18.6 dB [8]. There is also a clear trade-off between photon loss and squeezing. If the hardware has higher photon loss, the squeezing must be even higher to compensate.

These results are important because they provide a concrete engineering target. While 19 dB of squeezing is difficult to achieve with current technology, the existence of a finite threshold proves that fault-tolerant bosonic computing is possible. If these targets can be reached, the surface–GKP code should in principle allow logical errors to be suppressed to arbitrarily low levels using only a manageable number of physical modes.

5.6 Towards an optical surface–GKP code

Experimental realisation of a surface–GKP code remains a major challenge, but as discussed in §4.3, key components have recently been demonstrated. Optical GKP states have been generated on integrated photonic chips using squeezed light and heralded measurements,⁹ exhibiting the lattice structure and non-Gaussian features required for GKP encoding [23]. These results suggest that scalable, chip-based photonic platforms can produce approximate grid states suitable for bosonic encoding.

However, the achieved squeezing is still below the thresholds discussed in §5.5, and optical loss remains a dominant limitation [23]. While single-mode GKP states are now experimentally accessible in optics, the multi-mode entangling operations and repeated stabiliser measurements required for a full surface–GKP implementation have not yet been realised.

Architectural proposals aimed at overcoming these limitations emphasise static linear-optical networks and cluster-state constructions that avoid inline squeezing and active reconfiguration of optics, thereby reducing accumulated loss and hardware complexity [9, 29]. Together, these developments outline a plausible route toward optical surface–GKP codes, contingent on further improvements in squeezing and loss reduction.

⁸The squeezing values quoted in §5.5 follow the convention in [8], where the squeezing is defined in decibels as $S = -10\log_{10}(2\sigma^2)$.

⁹A heralded measurement is where a specific measurement outcome on an ancillary mode signals (heralds) the successful preparation of the desired non-Gaussian state in the other mode.

6 Conclusion

The Gottesman–Kitaev–Preskill code has evolved from a theoretical proposal into an experimentally relevant component of bosonic quantum error correction. By leveraging the infinite-dimensional Hilbert space of a harmonic oscillator, GKP states provide a hardware-efficient encoding that can reduce the number of physical modes per logical qubit compared to purely qubit-based implementations. This advantage becomes particularly clear in concatenated architectures. For comparable logical error rates, a surface–GKP code can significantly reduce the number of physical components required for fault-tolerant processors [9].

Experimental progress has substantially improved the outlook for GKP-based computing. Once considered unrealistic due to demanding squeezing requirements, GKP states have now been prepared and stabilised across trapped-ion, superconducting circuit-QED, and photonic platforms [19, 20, 23]. Advances in squeezing technology and real-time classical control have been central to this progress. In particular, demonstrating repeated, measurement-based feedback confirms that GKP qubits can be actively stabilised in realistic hardware, a necessary precondition for any concatenated fault-tolerant scheme.

The main challenge now lies in moving from single-mode state preparation to multi-mode entangled architectures with repeated stabiliser measurements. On the theory side, soft-decision decoding offers improved performance by exploiting the analog information in GKP measurement outcomes [8, 28]. On the hardware side, the primary bottlenecks remain achieving squeezing levels approaching the circuit-level threshold of approximately 18.6 dB [8] and developing high-fidelity entangling gates that do not introduce significant photon loss.

Overall, the GKP code can be viewed as a bridge between CV systems and standard qubit stabiliser theory, leveraging the natural structure of the underlying oscillator physics rather than abstracting it away. As the field progresses toward large-scale fault tolerance, integrating GKP states into concatenated architectures represents a promising and increasingly realistic route toward robust, scalable quantum computing.

References

- [1] P. Shor, “Algorithms for quantum computation: discrete logarithms and factoring”, in [Proceedings 35th annual symposium on foundations of computer science](#) (1994), pp. 124–134.
- [2] L. K. Grover, “A fast quantum mechanical algorithm for database search”, in [Proceedings of the twenty-eighth annual acm symposium on theory of computing](#), STOC ’96 (1996), pp. 212–219.
- [3] S. Lloyd, “Universal quantum simulators”, [Science](#) **273**, 1073–1078 (1996).
- [4] D. Aharonov and M. Ben-Or, “Fault-tolerant quantum computation with constant error rate”, (1999), [arXiv:quant-ph/9906129 \[quant-ph\]](#).
- [5] A. G. Fowler, M. Mariantoni, J. M. Martinis and A. N. Cleland, “Surface codes: towards practical large-scale quantum computation”, [Phys. Rev. A](#) **86**, 032324 (2012).
- [6] D. Gottesman, A. Kitaev and J. Preskill, “Encoding a qubit in an oscillator”, [Phys. Rev. A](#) **64**, 012310 (2001).
- [7] A. L. Grimsmo and S. Puri, “Quantum error correction with the Gottesman–Kitaev–Preskill code”, [PRX Quantum](#) **2**, 020101 (2021).
- [8] K. Noh and C. Chamberland, “Fault-tolerant bosonic quantum error correction with the surface–Gottesman–Kitaev–Preskill code”, [Phys. Rev. A](#) **101**, 012316 (2020).
- [9] B. M. Terhal, J. Conrad and C. Vuillot, “Towards scalable bosonic quantum error correction”, [Quantum Sci. Technol.](#) **5**, 043001 (2020).
- [10] A. L. Grimsmo, J. Combes and B. Q. Baragiola, “Quantum computing with rotation-symmetric bosonic codes”, [Phys. Rev. X](#) **10**, 011058 (2020).

- [11] X. Yang, S. Zamora, R. Chaves, U. L. Andersen, J. B. Brask and A. de Oliveira Junior, “Robust bell nonlocality from gottesman-kitaev-preskill states”, (2026), [arXiv:2601.16189 \[quant-ph\]](#).
- [12] S. Glancy and E. Knill, “Error analysis for encoding a qubit in an oscillator”, *Phys. Rev. A* **73**, 012325 (2006).
- [13] A. M. Steane, “Active stabilization, quantum computation, and quantum state synthesis”, *Phys. Rev. Lett.* **78**, 2252–2255 (1997).
- [14] E. Knill, “Quantum computing with realistically noisy devices”, *Nature* **434**, 39–44 (2005).
- [15] S. D. Bartlett and B. C. Sanders, “Universal continuous-variable quantum computation: requirement of optical nonlinearity for photon counting”, *Phys. Rev. A* **65**, 042304 (2002).
- [16] A. Mari and J. Eisert, “Positive wigner functions render classical simulation of quantum computation efficient”, *Phys. Rev. Lett.* **109**, 230503 (2012).
- [17] J. Hastrup, M. V. Larsen, J. S. Neergaard-Nielsen, N. C. Menicucci and U. L. Andersen, “Unsuitability of cubic phase gates for non-clifford operations on gottesman-kitaev-preskill states”, *Phys. Rev. A* **103**, 032409 (2021).
- [18] S. Bravyi and A. Kitaev, “Universal quantum computation with ideal clifford gates and noisy ancillas”, *Phys. Rev. A* **71**, 022316 (2005).
- [19] C. Flühmann, T. L. Nguyen, M. Marinelli, V. Negnevitsky, K. Mehta and J. P. Home, “Encoding a qubit in a trapped-ion mechanical oscillator”, *Nature* **566**, 513–517 (2019).
- [20] P. Campagne-Ibarcq et al., “Quantum error correction of a qubit encoded in grid states of an oscillator”, *Nature* **584**, 368–372 (2020).
- [21] J. L. O’Brien, A. Furusawa and J. Vučković, “Photonic quantum technologies”, *Nature Photonics* **3**, 687–695 (2009).
- [22] C. Weedbrook et al., “Gaussian quantum information”, *Rev. Mod. Phys.* **84**, 621–669 (2012).
- [23] M. V. Larsen et al., “Integrated photonic source of gottesman-kitaev-preskill qubits”, *Nature* **642**, 587–591 (2025).
- [24] S. Konno et al., “Logical states for fault-tolerant quantum computation with propagating light”, *Science* **383**, 289–293 (2024).
- [25] M. Caleffi, M. Amoretti, D. Ferrari, J. Illiano, A. Manzalini and A. S. Cacciapuoti, “Distributed quantum computing: a survey”, *Computer Networks* **254**, 110672 (2024).
- [26] PsiQuantum Team, “A manufacturable platform for photonic quantum computing”, *Nature* **641**, 876–883 (2025).
- [27] D. Gottesman, “Stabilizer codes and quantum error correction”, Ph.D. thesis (Caltech, May 1997), [arXiv:quant-ph/9705052 \[quant-ph\]](#).
- [28] K. Fukui, A. Tomita, A. Okamoto and K. Fujii, “High-threshold fault-tolerant quantum computation with analog quantum error correction”, *Phys. Rev. X* **8**, 021054 (2018).
- [29] I. Tzitrin et al., “Fault-tolerant quantum computation with static linear optics”, *PRX Quantum* **2**, 040353 (2021).